Multimodal Learning for Medicine & Healthcare: The Challenges and Opportunities (IN2107)

Marta Hasny & Laura Daza & Jun Li
Chair for Computational Imaging and AI in Medicine
Winter Semester 2026/2027

About Us

Chair for Computational Imaging and AI in Medicine
Prof. Dr. Julia Schnabel

About Us

Dr. Laura Daza

- Medical image segmentation
- Foundation models
- Multimodal learning

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Marta Hasny

- Foundation models
- Multimodal learning
- Cardiology

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Jun Li

- Large Vision Language Model
- Abnormality Grounding
- Multimodal Learning

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1) Organization

Schedule (Subject to Change)

Week	Session
Week 1	Introduction to the Seminar
Week 2	How to read a paper and do poster presentations
Week 3	Self-supervised Learning
Week 4	Challenges of Multimodal Learning in Medicine
Week 5	Student Presentations
Week 6	Student Presentations
Week 7	Student Presentations
Week 8	Student Presentations
Week 9	Invited Talk
Week 10	Multimodal Foundation Models
Week 11	Student Presentations
Week 12	Student Presentations
Week 14	Invited Talk
Week 15	**Poster Session**

When?

Thursdays: 2-4pm (subject to room availability)

Deliverables & grading

Oral presentations
- 30 minutes presentation + 10 minutes questions
  - About your selected paper + small review of the topic
- Presentation date depends on the topic

Poster presentations
- Scientific poster
  - About your selected paper only
- 5 minutes presentation

Paper selection
- We will provide a list of papers
- We will assign the papers based on your submitted selections
- Papers on both multimodal methodologies and medical applications

Grading
Oral + poster presentations + attendance + participation

Poster session at end of semester

Paper Topics from Previous Semesters

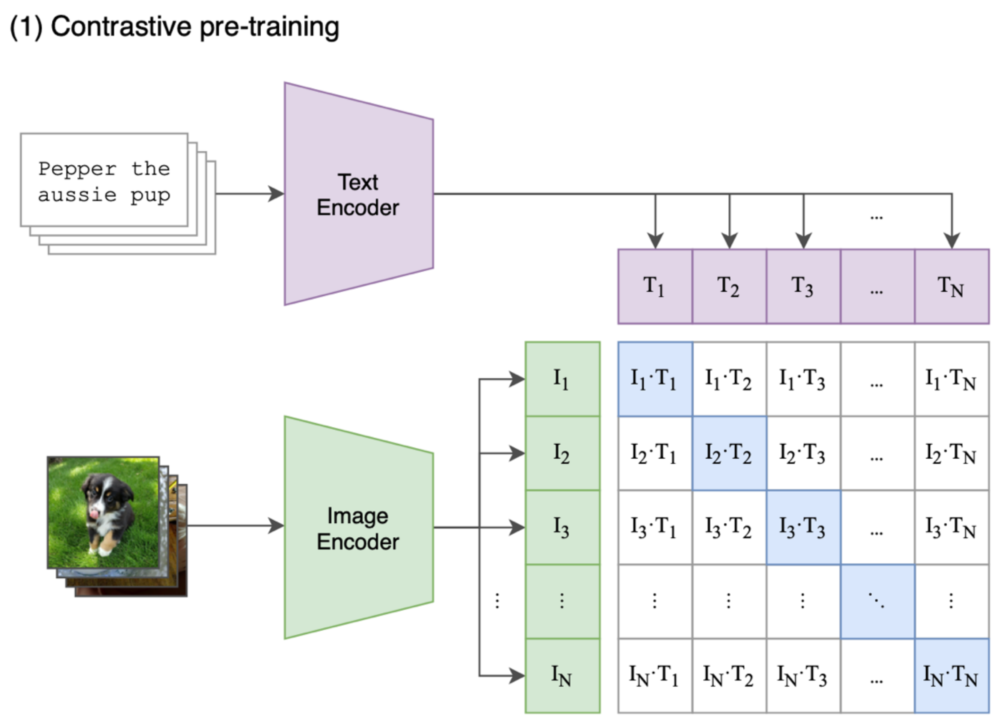
- Multimodal Fusion
- Addressing the Challenges of Multimodal Learning
- Multimodal Pretraining
- Multimodal Medical Foundation Models
- Multimodal Generalization & Representation

We try to select recent papers while providing the background to better understand them in the lectures. That way you can have an overview of what is going on in the field right now ☺

2) Multimodal learning
What will the seminar be about?

Multimodal Learning

Vision-Language Models



[1]

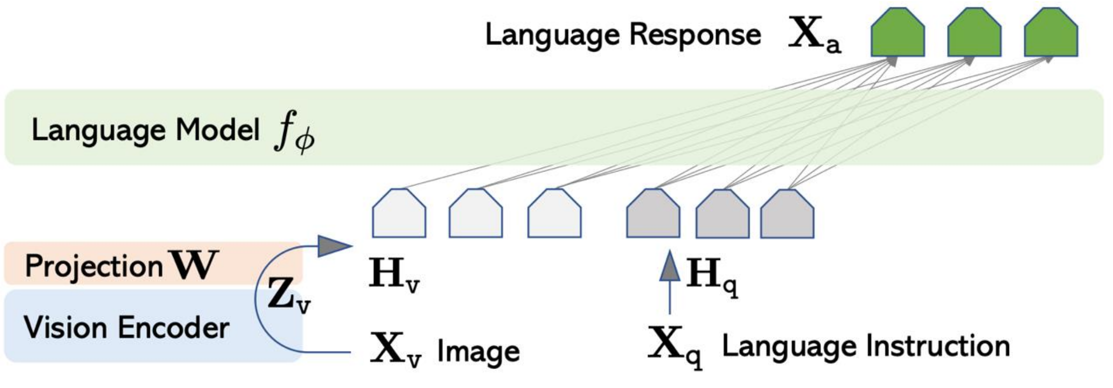


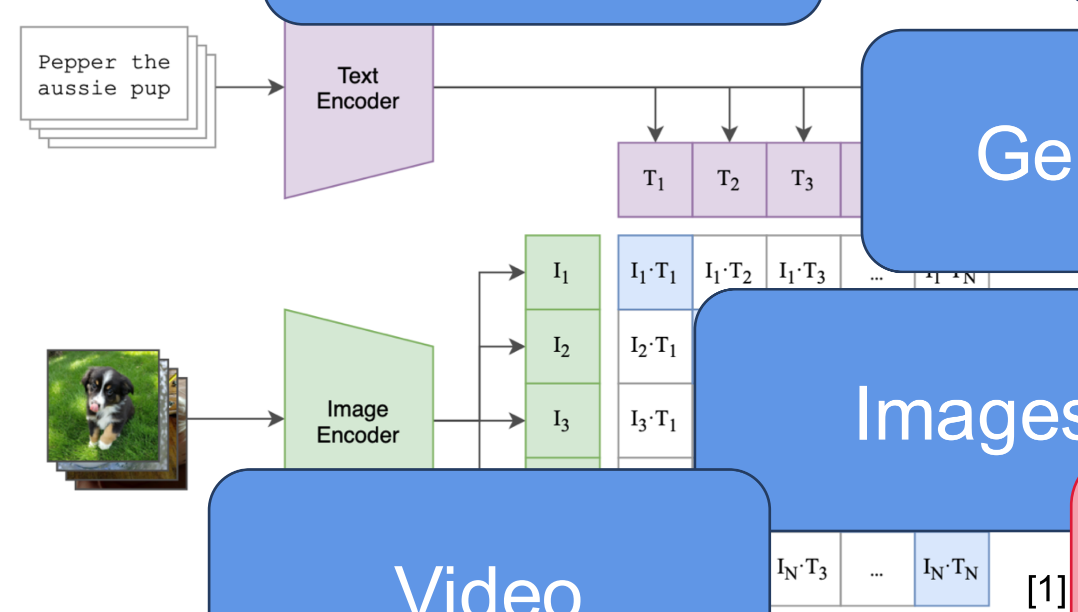
Figure 1: LLaVA network architecture.

[2]

[1] Radford, Alec, et al. "Learning transferable visual models from natural language supervision." *International conference on machine learning*. PmLR, 2021.
[2] Liu, Haotian, et al. "Visual instruction tuning." *Advances in neural information processing systems* 36 (2023): 34892-34916.

Multimodal Learning

Vision-Lan…



[1]

[2]

Text

Tabular Data

Genomics

Audio

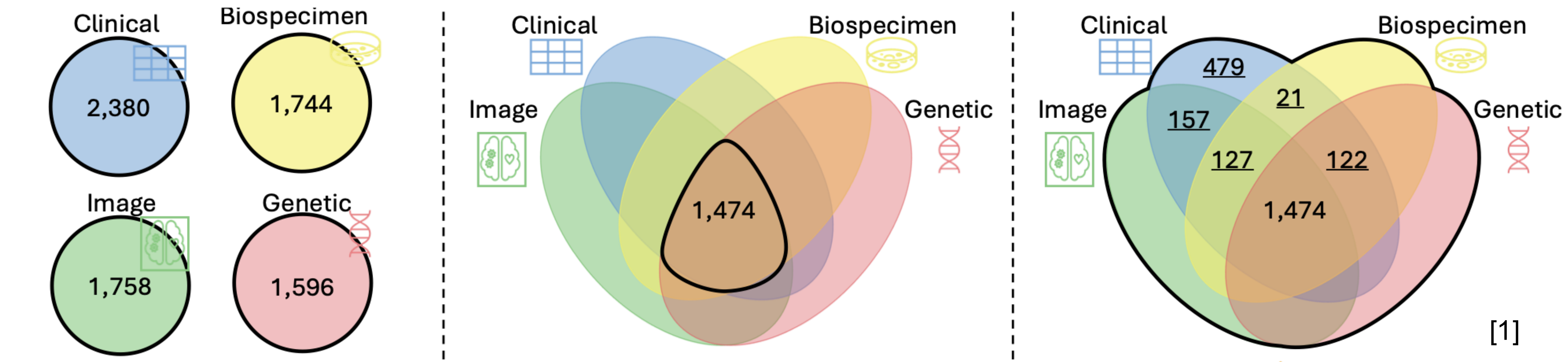
Images

Video

Different Imaging Modalities

[1] Radford, Alec, et al. "Learning transferable visual models from natural language supervision." *International con…*
[2] Liu, Haotian, et al. "Visual instruction tuning." *Advances in neural information processing systems* 36 (2023): 34892-…

Multimodal Learning



(a) Existing Works – Single-modal (b) Existing Works – Multi-modal (c) Ours

Multimodal methods often assume that all modalities are available → But in clinical practice that's not the case…

[1] Yun, Sukwon, et al. "Flex-moe: Modeling arbitrary modality combination via the flexible mixture-of-experts." Advances in Neural Information Processing Systems 37 (2024): 98782-98805.

Have a nice day!

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